

2.6 Stationary Infinite-Horizon Problems

The finite-horizon problems studied so far have a terminal stage and generally use stage-dependent dynamics, costs, and feedback laws. This section changes both features: the horizon is indefinite, and the ingredients of the decision problem are stationary. An infinite horizon alone does not imply stationarity. Here, *stationarity* means that the dynamics, one-stage cost, admissible-control set, and disturbance law do not change with the absolute stage. Thus the stationary counterparts of the earlier models are

$$x_{k+1} = f(x_k, u_k), \quad g_k(x_k, u_k) = g(x_k, u_k), \quad \text{deterministic,} \quad (2.55)$$

$$x_{k+1} = f(x_k, u_k, w_k), \quad g_k(x_k, u_k, w_k) = g(x_k, u_k, w_k), \quad \text{stochastic,} \quad (2.56)$$

with $u_k \in \mathcal{U}(x_k)$. In the stochastic model, $P^w(\cdot | x, u)$ denotes the stage-independent conditional law of w_k ; it is the stationary counterpart of $P_k^w(\cdot | x_k, u_k)$ in Section 2.3. Expectations written as $\mathbb{E}_{w|x,u}[\cdot]$ below are taken under this law.

Let $\mathcal{M}_\infty$ denote the admissible deterministic Markov feedback sequences $\mu = (\mu_0, \mu_1, \dots)$ satisfying $\mu_k(x) \in \mathcal{U}(x)$ for every state and stage. We first consider the *discounted infinite-horizon problem*. Let $0 < \alpha < 1$ denote the *discount factor*; the multiplier α^k assigns progressively less weight to costs incurred farther in the future. The discounted infinite-horizon cost of μ is

$$J_\mu(x) = \lim_{N \rightarrow \infty} \mathbb{E}^\mu \left[\sum_{k=0}^{N-1} \alpha^k g(x_k, \mu_k(x_k), w_k) \mid x_0 = x \right], \quad 0 < \alpha < 1, \quad (2.57)$$

where w_k is omitted in the deterministic case. Thus Equation (2.57) defines the discounted infinite-horizon criterion. The optimal value is $J^*(x) = \inf_{\mu \in \mathcal{M}_\infty} J_\mu(x)$. The infimum is used at this general definition because an optimizing policy has not yet been shown to exist. For the finite-state, finite-control reference model adopted below, an optimum is attained, and the Bellman equations and exact tabular algorithms therefore use a minimum. A stationary policy uses one rule at every stage, $u_k = \mu(x_k)$, and its value is denoted J_μ . This notation initially presumes that the limit exists. Under the bounded-cost discounted assumption stated later in this section—in particular, $|g(x, u, w)| \leq g_{\max} < \infty$ with probability one—

$$\sum_{k=0}^{\infty} \alpha^k |g(x_k, \mu_k(x_k), w_k)| \leq \frac{g_{\max}}{1 - \alpha} \quad \text{with probability one.}$$

Consequently, the sample-path series converges absolutely and the expected cost in Equation (2.57) is finite. The same discount factor also creates the contraction property used to justify VI and PI. Discounting is therefore both a modeling choice and a convenient reference theory. If costs far in the future should not be attenuated, a different objective is required. The SSP formulation introduced below provides one such alternative for terminating tasks.

2.6.1 Why One Value Function Can Be Reused

In a finite-horizon problem, $J_k^*(x)$ depends on k because the remaining number of decisions, model, and terminal boundary depend on the stage. In a stationary infinite-horizon problem,

however, restarting from the same state presents the same remaining optimization problem at every absolute stage. The tail problem from state x is therefore the original infinite-horizon problem restarted at x , not a new stage-dependent problem. Its reusable objects are written

$$J^*(x), \quad J_\mu(x), \quad Q^*(x, u), \quad Q_\mu(x, u), \quad (2.58)$$

without a physical-time subscript. For a discounted objective, the tail that begins at physical stage k carries the common positive factor α^k . Removing this factor does not change which continuation is minimizing, so the restarted problem may begin its discount exponent at zero.

If calendar time, route position, battery age, or another exogenous phase changes the future problem, that variable must instead be included in the state. Section 2.9 returns to this point through a finite-to-infinite-horizon reformulation that includes the stage in the state when it is needed. Figure 2.11 summarizes the stationary case, in which the same model, value function, and policy can be reused whenever the same state recurs.

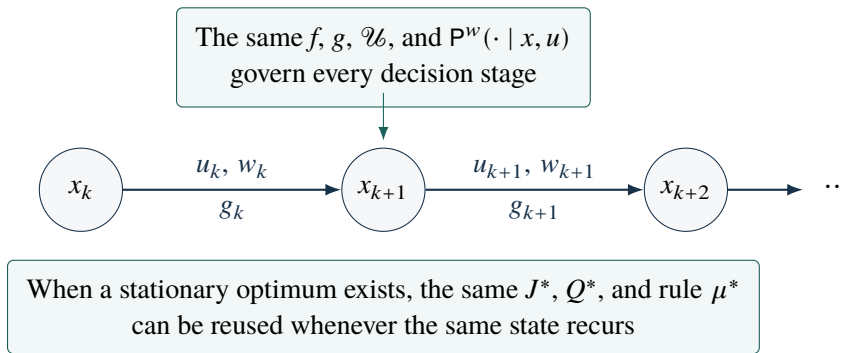


Figure 2.11: Stationarity in an infinite-horizon problem. Here $g_j = g(x_j, u_j, w_j)$. The stage number advances, but the decision model and the value objects do not change with the absolute stage.

A preview of infinite-horizon theory.

Before treating VI as an algorithm, it is helpful to reinterpret finite-horizon backward DP for a stationary model. For an N -stage truncation, let $V_{k|N}$ denote the normalized discounted cost-to-go from stage k , with $V_{N|N} = 0$ and the discount exponent restarted at the current stage. Backward DP computes $V_{k|N}$ from $V_{k+1|N}$. Setting $i = N - k$ and $J_i := V_{N-i|N}$ reverses this indexing: J_i is the optimal cost of the same stationary model truncated after i decisions. Here i is a horizon-length and algorithm-iteration index, not physical time. Because the normalized stationary recursion and zero terminal cost do not depend on the chosen terminal label N , the resulting J_i is the same for every truncation with $N \geq i$. Figure 2.12 shows the correspondence.

The common recursion is

$$\begin{aligned} J_0(x) &= 0, \\ J_{i+1}(x) &= \min_{u \in \mathcal{U}(x)} \mathbb{E}_{w|x, u} [g(x, u, w) + \alpha J_i(f(x, u, w))], \quad \forall x \in \mathcal{X}, \quad i = 0, 1, \dots \end{aligned} \quad (2.59)$$

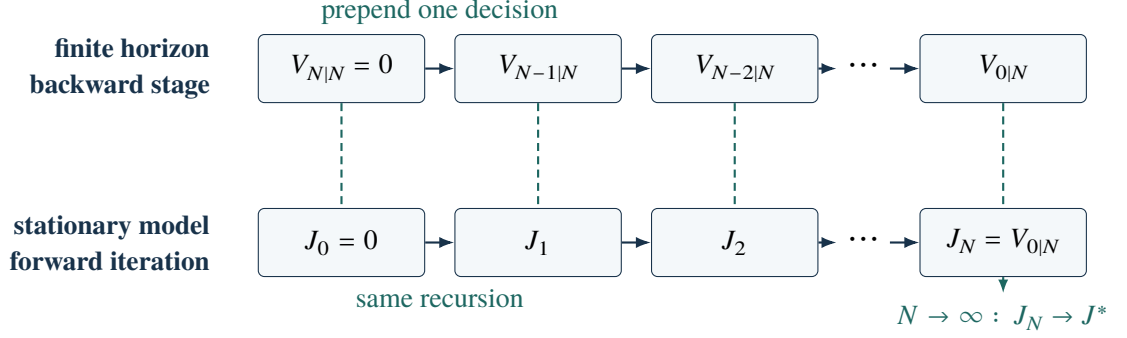


Figure 2.12: Finite-horizon backward DP and its stationary horizon-length reinterpretation. With stationary dynamics and costs and zero terminal cost, $J_i = V_{N-i|N}$ represents an i -decision tail problem; for a fixed N , $J_N = V_{0|N}$, and the sequence converges to J^* as the horizon grows.

$J_0 = 0$ represents a problem with no remaining decisions. One update adds one decision in front of that zero-cost tail; each subsequent update adds one more decision. Because the model is stationary, the right-hand computation is the same at every update. When the same recursion is continued as an algorithm, it is called *value iteration* (VI). Under the finite, bounded-cost, discounted assumptions stated below, Section 2.7 shows that

$$\lim_{i \rightarrow \infty} \|J_i - J^*\|_\infty = 0, \quad J^*(x) = \lim_{i \rightarrow \infty} J_i(x), \quad \forall x \in \mathcal{X}. \quad (2.60)$$

The limit is characterized by the discounted stationary Bellman optimality equation

$$J^*(x) = \min_{u \in \mathcal{U}(x)} \mathbb{E}_{w|x,u} [g(x, u, w) + \alpha J^*(f(x, u, w))], \quad \forall x \in \mathcal{X}. \quad (2.61)$$

In the finite-state, finite-control reference model, Equation (2.61) is a coupled system containing one equation for every state, not a relation imposed only along one realized trajectory. A minimizing control at every state defines the stationary optimal policy established in the next section. Thus Equation (2.59) is simultaneously finite-horizon backward DP with reversed indexing and zero-initialized VI. This is the preview used in Bertsekas (2019, Sec. 4.1). Section 2.9 develops a different construction: it exactly embeds a given finite-horizon problem in a stage-augmented infinite-horizon model.

2.6.2 A Second Infinite-Horizon Model: Stochastic Shortest Path

The objective in Equation (2.57) and the Bellman equation in Equation (2.61) define the discounted infinite-horizon model. Discounting is not the only useful infinite-horizon criterion. A *stochastic shortest path* (SSP) problem is an undiscounted total-cost problem that ends when it reaches a distinguished absorbing terminal state, denoted here by Δ , that incurs no further cost:

$$f(\Delta, u, w) = \Delta, \quad g(\Delta, u, w) = 0. \quad (2.62)$$

With the hitting time $\tau = \inf\{k : x_k = \Delta\}$, its undiscounted objective is

$$J_\mu(x) = \mathbb{E}^\mu \left[\sum_{k=0}^{\tau-1} g(x_k, \mu_k(x_k), w_k) \mid x_0 = x \right]. \quad (2.63)$$

After Δ is reached, there is no further physical decision to make. A policy is called *proper* when its hitting time satisfies

$$\Pr^\mu\{\tau < \infty \mid x_0 = x\} = 1 \quad \forall x \in \mathcal{X} \setminus \{\Delta\}.$$

Here $\Pr^\mu$ denotes probability under the closed-loop law induced by μ ; the superscript is therefore not an exponent. A proper policy reaches the terminal state in a finite, possibly random and policy-dependent number of stages with probability one. For a general SSP, properness alone is not sufficient: additional cost and termination conditions may be needed to make the optimal cost well defined and to ensure convergence of DP algorithms (Bertsekas 2019, Sec. 4.2). This chapter does not develop that general theory. Deterministic shortest-path problems fit the SSP model, and many episodic control problems do as well when the state includes any phase or remaining-task information needed for the Markov property.

The discounted and SSP equations look similar, but their elementary theories are not interchangeable. For SSP, $\alpha = 1$, so the ordinary discounted sup-norm contraction used below is generally unavailable. Convergence and well-posedness instead depend on the SSP's cost and termination structure. Here SSP is used only for the stage-augmented finite problem in Section 2.9. That construction is acyclic, has finite one-stage costs, and, starting from layer k , reaches Δ after exactly $N - k$ genuine control decisions and one terminal transition under every policy. Its VI and PI properties can therefore be established directly. Broader SSP theory is outside the present scope.

2.6.3 Finite and Continuous State–Control Spaces

Equations (2.55) and (2.56) allow finite or continuous states and controls. The Bellman equation can be written in either setting. What changes is how the value function, expectation, and feasible-control minimization are represented and computed. The exact guarantees in the next section are deliberately stated for a finite-state, finite-control reference model so that the central VI and PI logic is visible without a general-space detour.

Finite spaces.

When $\mathcal{X}$ and every $\mathcal{U}(x)$ are finite, a value can be stored as a table, an expectation can be written as a probability-weighted sum, and a minimum exists by enumeration. A tabular value representation stores one scalar $J(x)$ for every state; a tabular Q representation stores one scalar $Q(x, u)$ for every feasible state–control pair. An *exact tabular backup* evaluates one represented entry using the declared costs and transition probabilities without sampling, function fitting, approximate search, or early solver termination. An exact synchronous sweep performs such a backup at every represented entry. Here, exactness is relative to the declared finite-state,

finite-control numerical model, not a claim that the table reproduces the underlying physical system exactly. This qualification is retained whenever the chapter reports an “exact” numerical result.

The following assumption fixes that finite-state, finite-control reference model. In the terminology introduced in Section 2.1, it is a finite Markov decision process (MDP).

Assumption 2.3 (Discounted finite-state, finite-control reference model). The state set $\mathcal{X}$ and every admissible-control set $\mathcal{U}(x)$ are finite, and every $\mathcal{U}(x)$ is nonempty. For every $x \in \mathcal{X}$ and $u \in \mathcal{U}(x)$, the successor $f(x, u, w)$ lies in $\mathcal{X}$ and $|g(x, u, w)| \leq g_{\max} < \infty$ with probability one under $\mathbf{P}^w(\cdot \mid x, u)$. The model and disturbance law are stationary, and $0 < \alpha < 1$.

Finiteness of $\mathcal{X}$ only implies that every finite-valued table has a finite sup norm; it does not by itself make an infinite-horizon policy cost finite. The bounded one-stage cost and $0 < \alpha < 1$ instead give $\|J_\mu\|_\infty \leq g_{\max}/(1 - \alpha)$ for every admissible policy and the same bound for J^* .

Continuous spaces.

The same structural equations use functions, integrals or samples, and an optimization over continuous controls. If existence of a minimizing feedback law has not been established, the mathematically safe notation is an infimum rather than a minimum. Additional compactness, continuity, and measurability conditions are then needed to guarantee that this infimum is attained and defines an admissible feedback law. The finite-table guarantees below must not be transferred automatically to every continuous-space model; in particular, finite termination of PI is a finite-policy conclusion.

If a continuous steering or torque command is replaced by a finite control library, control discretization has entered the numerical model. It is not mandatory: LQR can minimize analytically, MPC can optimize over continuous inputs, and a parameterized actor can output a continuous control. Likewise, a continuous value cannot generally be stored as a complete table and therefore uses analytic structure, interpolation, or approximation.

The slalom-tracking example later in this chapter makes this distinction concrete. Physical lateral motion is continuous, but the declared teaching MDP uses 65 two-state grid points and three steering controls. Exact VI and PI are exact for that finite grid. Applying the same Bellman logic to the underlying continuous problem requires an analytic representation, continuous optimization, discretization, or approximation.

2.7 Exact Infinite-Horizon DP: VI and PI

Section 2.6 introduced the stationary discounted problem and its Bellman optimality equation. This section introduces the two principal exact DP algorithms for solving that equation on the declared finite-state, finite-control reference model. Recall from Section 2.2 that a *Bellman backup* evaluates a Bellman recursion at one state. When the backup minimizes over all feasible current controls, it is a *Bellman optimality backup*; applying that backup at every state is one *VI sweep*. By contrast, fixing the control to a policy gives a *policy-evaluation backup*. A *Bellman*

fixed point is a value table reproduced without change by the corresponding full sweep. *Value iteration* (VI) repeats VI sweeps whose iterates converge to the optimal fixed point. *Policy iteration* (PI) instead alternates two tasks: it evaluates the infinite-horizon cost of the current policy and then improves that policy by a greedy Bellman decision. The operator notation introduced next compresses these repeated computations; it is not a separate algorithm.

2.7.1 From the Bellman Equation to Operators

Equation (2.61) gives the Bellman optimality equation that VI and PI must solve. Under assumption 2.3, any value table is a vector in $\mathbb{R}^{|\mathcal{X}|}$. For such a table J , define the Bellman optimality operator T and the *fixed-policy Bellman operator* T_μ , used for policy evaluation—the DP operator corresponding to a fixed policy μ —directly from Equation (2.56):

$$(TJ)(x) = \min_{u \in \mathcal{U}(x)} \mathbb{E}_{w|x,u} [g(x, u, w) + \alpha J(f(x, u, w))], \quad \forall x \in \mathcal{X}, \quad (2.64)$$

$$(T_\mu J)(x) = \mathbb{E}_{w|x, \mu(x)} [g(x, \mu(x), w) + \alpha J(f(x, \mu(x), w))], \quad \forall x \in \mathcal{X}. \quad (2.65)$$

The deterministic forms remove the expectation and w . The first operator compares all feasible current controls. The second fixes $u = \mu(x)$ and therefore evaluates rather than improves the policy.

For any J , its one-step Q-factor is

$$Q_J(x, u) = \mathbb{E}_{w|x,u} [g(x, u, w) + \alpha J(f(x, u, w))], \quad \forall x \in \mathcal{X}, \quad u \in \mathcal{U}(x). \quad (2.66)$$

Thus $(TJ)(x) = \min_{u \in \mathcal{U}(x)} Q_J(x, u)$. The value $J(x)$ evaluates a state before the present control is selected; $Q_J(x, u)$ evaluates committing to u now and using J for the continuation. At the optimum,

$$J^*(x) = \min_{u \in \mathcal{U}(x)} Q^*(x, u), \quad Q^*(x, u) = Q_{J^*}(x, u), \quad \forall x \in \mathcal{X}, \quad u \in \mathcal{U}(x). \quad (2.67)$$

Finite transition-table notation.

For the finite-state, finite-control reference model, define

$$p(x' | x, u) = \Pr\{f(x, u, w) = x' | x, u\}, \quad \bar{g}(x, u) = \mathbb{E}_{w|x,u} [g(x, u, w)]. \quad (2.68)$$

Then Equation (2.64) becomes

$$(TJ)(x) = \min_{u \in \mathcal{U}(x)} \left\{ \bar{g}(x, u) + \alpha \sum_{x' \in \mathcal{X}} p(x' | x, u) J(x') \right\}, \quad \forall x \in \mathcal{X}. \quad (2.69)$$

This transition notation is convenient for matrices and tables; it is a finite-space representation of the x, u, w expectation, not a different control problem.

Whether it is written in expectation form or in the equivalent finite transition-table form, the Bellman optimality equation is the fixed-point equation $J^* = TJ^*$. Likewise, a fixed policy satisfies $J_\mu = T_\mu J_\mu$. The two algorithms below use these equations in different ways.

2.7.2 Exact VI

VI solves the Bellman optimality equation (2.61) by repeatedly applying its right-hand side:

$$J_{i+1}(x) = (TJ_i)(x), \quad \forall x \in \mathcal{X}, \quad i = 0, 1, \dots \quad (2.70)$$

Written in the x, u, w notation, its deterministic and stochastic forms are

$$J_{i+1}(x) = \min_{u \in \mathcal{U}(x)} \{g(x, u) + \alpha J_i(f(x, u))\}, \quad \forall x \in \mathcal{X}, \quad \text{deterministic}, \quad (2.71)$$

$$J_{i+1}(x) = \min_{u \in \mathcal{U}(x)} \mathbb{E}_{w|x, u} [g(x, u, w) + \alpha J_i(f(x, u, w))], \quad \forall x \in \mathcal{X}, \quad \text{stochastic}. \quad (2.72)$$

Here i is an algorithm iteration, not physical time. One synchronous iteration evaluates the right-hand side for every $x \in \mathcal{X}$. Thus VI solves a full-state Bellman system; it is not a simulation along one trajectory.

The central discounted finite-state, finite-control result records what this iteration computes, what equation its limit solves, and how that value determines an optimal policy. It also gives the corresponding VI for a fixed policy (Bertsekas 2019, Props. 4.3.1–4.3.4).

Theorem 2.1 (VI convergence, Bellman equation, and optimality). *Under assumption 2.3, the following statements hold.*

1. *For any initial value table J_0 , the VI sequence in Equation (2.70) converges to J^* : $\lim_{i \rightarrow \infty} \|J_i - J^*\|_\infty = 0$.*
2. *The optimal value J^* is the unique solution of the Bellman equation $J = TJ$.*
3. *For any stationary policy μ and any initial table V_0 , the fixed-policy VI, also called iterative policy evaluation,*

$$V_{m+1}(x) = (T_\mu V_m)(x), \quad \forall x \in \mathcal{X}, \quad m = 0, 1, \dots,$$

converges to J_μ , the unique solution of $J = T_\mu J$.

4. *A deterministic stationary policy μ^* is optimal if and only if, for every $x \in \mathcal{X}$, it attains the minimum in the Bellman equation, equivalently*

$$T_{\mu^*} J^* = TJ^* = J^*. \quad (2.73)$$

The theorem states the main conclusion. The monotonicity and contraction properties below provide a compact explanation of convergence and a useful rate bound.

Two properties explain why the repeated backups work. First, *monotonicity* means that componentwise ordering is preserved:

$$J \leq J' \implies TJ \leq TJ', \quad T_\mu J \leq T_\mu J'. \quad (2.74)$$

Second, define the sup norm of a value table by

$$\|J\|_\infty = \max_{x \in \mathcal{X}} |J(x)|. \quad (2.75)$$

The following operator properties explain the theorem above and provide the rate bound below; compare Bertsekas (2019, Prop. 4.3.5).

Proposition 2.4 (Monotonicity and contraction of discounted Bellman operators). *Under assumption 2.3, T and every T_μ are monotone and, for all value tables J, J' and every admissible stationary policy μ , satisfy*

$$\|TJ - TJ'\|_\infty \leq \alpha \|J - J'\|_\infty, \quad \|T_\mu J - T_\mu J'\|_\infty \leq \alpha \|J - J'\|_\infty. \quad (2.76)$$

Proof. If $J \leq J'$ componentwise, expectation and minimization preserve the order, which proves Equation (2.74). For contraction, the elementary inequality $|\min_u a_u - \min_u b_u| \leq \max_u |a_u - b_u|$ gives, for every state,

$$|(TJ)(x) - (TJ')(x)| \leq \alpha \max_{u \in \mathcal{U}(x)} \mathbb{E}_{w|x,u} [|J(f(x, u, w)) - J'(f(x, u, w))|] \leq \alpha \max_{y \in \mathcal{X}} |J(y) - J'(y)|.$$

This statewise inequality holds for every x . Taking the maximum over x and using $\max_y |J(y) - J'(y)| = \|J - J'\|_\infty$ proves the first contraction bound; fixing the control to $\mu(x)$ gives the same argument for the second. $\square$

The contraction mapping theorem gives a unique fixed point of T and of every T_μ . The standard discounted DP results identify these fixed points as J^* and J_μ , respectively (Bertsekas 2019, Props. 4.3.2–4.3.3). Repeated application of the contraction bound quantifies the VI convergence stated in theorem 2.1 (Bertsekas 2019, Props. 4.3.1 and 4.3.5):

$$\|J_i - J^*\|_\infty \leq \alpha^i \|J_0 - J^*\|_\infty. \quad (2.77)$$

Hence every initial value table converges to the unique solution J^* . The unknown J^* makes Equation (2.77) a theoretical rate rather than a stopping test. A computable alternative is the Bellman residual, for which the same contraction argument gives

$$\|J - J^*\|_\infty \leq \frac{1}{1 - \alpha} \|TJ - J\|_\infty. \quad (2.78)$$

This residual bound is a manuscript corollary rather than a separately quoted result: apply the triangle inequality to $J - J^* = (J - TJ) + (TJ - TJ^*)$ and use contraction. It is computable only when the exact residual is evaluated over every state, and it bounds value error rather than directly bounding the performance loss of the greedy policy.

After any iterate, a greedy policy is obtained from

$$\mu_i(x) \in \arg \min_{u \in \mathcal{U}(x)} \mathbb{E}_{w|x,u} [g(x, u, w) + \alpha J_i(f(x, u, w))], \quad \forall x \in \mathcal{X}. \quad (2.79)$$

Equivalently, $T_{\mu_i} J_i = TJ_i = J_{i+1}$. The value sequence is guaranteed to converge. The greedy-policy sequence need not select one unchanging policy when several controls tie, but every policy greedy with respect to J^* is optimal by theorem 2.1. The convergence is asymptotic; “exact VI” means that its represented Bellman backups are exact, not that a finitely stopped iterate already equals J^* . Near $\alpha = 1$, the contraction can be slow.

Algorithm 2.3: Exact discounted VI

1. Choose any value table J_0 , a residual tolerance $\varepsilon > 0$, and set $i = 0$.
2. For every $x \in \mathcal{X}$, compute $B(x) = (TJ_i)(x)$, and set $r_i = \|B - J_i\|_\infty$.
3. If $r_i \leq \varepsilon$, return J_i and choose a greedy control $\mu_i(x)$ from Equation (2.79) for every $x \in \mathcal{X}$.
4. Otherwise set $J_{i+1} = B$, increment i , and repeat.

Q-factor VI and its sampled target.

The exact Q-factor form makes the VI correspondence explicit. For the stochastic discounted model,

$$Q^*(x, u) = \mathbb{E}_{w|x, u} \left[g(x, u, w) + \alpha \min_{v \in \mathcal{U}(f(x, u, w))} Q^*(f(x, u, w), v) \right], \quad \forall x \in \mathcal{X}, \quad u \in \mathcal{U}(x), \quad (2.80)$$

and the associated operator is

$$(T_Q Q)(x, u) = \mathbb{E}_{w|x, u} \left[g(x, u, w) + \alpha \min_{v \in \mathcal{U}(f(x, u, w))} Q(f(x, u, w), v) \right], \quad (2.81)$$

$$\forall x \in \mathcal{X}, \quad u \in \mathcal{U}(x).$$

Thus Q-factor VI is

$$Q_{i+1}(x, u) = (T_Q Q_i)(x, u), \quad \forall x \in \mathcal{X}, \quad u \in \mathcal{U}(x), \quad i = 0, 1, \dots \quad (2.82)$$

These are the system-and-disturbance forms of Bertsekas (2019, Eqs. (4.16)–(4.17)). Under assumption 2.3, the same minimum inequality used in proposition 2.4 gives, in the sup norm over feasible state–control pairs,

$$\|T_Q Q - T_Q Q'\|_\infty \leq \alpha \|Q - Q'\|_\infty.$$

Indeed, the inner minimum changes by at most the largest entrywise change in Q , and conditional expectation cannot enlarge this bound. Thus T_Q is an α -contraction, Q^* is its unique fixed point, and $Q_i \rightarrow Q^*$ from any finite-valued initial Q table. This contraction statement is a direct consequence of the displayed operator, rather than an additional quoted result.

The Q form also exposes the sample target that later becomes central to Q-learning. For a fixed (x, u) , hold the current Q fixed and draw a conditional transition sample with realized cost $g^{(s)} = g(x, u, w^{(s)})$ and successor $x^{+(s)} = f(x, u, w^{(s)})$. Then

$$Y^{(s)} = g^{(s)} + \alpha \min_{v \in \mathcal{U}(x^{+(s)})} Q(x^{+(s)}, v), \quad \mathbb{E}[Y^{(s)} | x, u] = (T_Q Q)(x, u). \quad (2.83)$$

One transition therefore estimates one Q-backup component. Chapter 4 turns this target into a complete Q-learning algorithm, including behavior-policy, coverage, and stepsize requirements; the update is not repeated here.

2.7.3 Exact PI

PI separates *policy evaluation* from *policy improvement*. Starting from a stationary policy μ_i , policy evaluation computes its infinite-horizon cost as the unique solution of

$$J_{\mu_i}(x) = \mathbb{E}_{w|x, \mu_i(x)} [g(x, \mu_i(x), w) + \alpha J_{\mu_i}(f(x, \mu_i(x), w))], \quad \forall x \in \mathcal{X}. \quad (2.84)$$

For deterministic dynamics, remove the expectation and w . In finite transition-probability notation, define $P_{\mu_i}(x, x') = p(x' | x, \mu_i(x))$ and $\bar{g}_{\mu_i}(x) = \bar{g}(x, \mu_i(x))$. Policy evaluation is then the linear system

$$(I - \alpha P_{\mu_i})J_{\mu_i} = \bar{g}_{\mu_i}. \quad (2.85)$$

Because P_{μ_i} is a row-stochastic matrix, $\|\alpha P_{\mu_i}\|_{\infty} = \alpha < 1$; consequently $I - \alpha P_{\mu_i}$ is nonsingular and the linear system has a unique solution. The same value can be obtained by VI for the fixed policy μ_i , equivalently by iterative *policy evaluation*. From any initial table $V_{i,0}$, compute

$$V_{i,m+1}(x) = (T_{\mu_i} V_{i,m})(x), \quad \forall x \in \mathcal{X}, \quad m = 0, 1, \dots, \quad V_{i,m} \rightarrow J_{\mu_i}. \quad (2.86)$$

Here m counts inner policy-evaluation sweeps, whereas i counts outer PI iterations. This is the policy-VI result in theorem 2.1. Exact PI either solves Equation (2.85) directly or continues the backups to their fixed point, conceptually $m \rightarrow \infty$. Stopping after a finite number m_i of sweeps gives modified or optimistic PI, introduced in Section 2.8.

Policy improvement then chooses a policy greedy with respect to the evaluated value:

$$\mu_{i+1}(x) \in \arg \min_{u \in \mathcal{U}(x)} \mathbb{E}_{w|x, u} [g(x, u, w) + \alpha J_{\mu_i}(f(x, u, w))], \quad \forall x \in \mathcal{X}. \quad (2.87)$$

A stationary policy is optimal if and only if it selects a minimizing control in $J^* = TJ^*$ at every state. PI approaches that condition by making the current policy greedy with respect to its own exactly evaluated value.

The next proposition gives the algorithm-level conclusion for the finite discounted model (Bertsekas 2019, Prop. 4.5.1).

Proposition 2.5 (Policy improvement and finite termination of exact PI). *Under assumption 2.3, evaluate a deterministic stationary policy μ_i exactly. If μ_i is not greedy with respect to J_{μ_i} , let μ_{i+1} be any policy that is greedy with respect to J_{μ_i} . Then*

$$J_{\mu_{i+1}}(x) \leq J_{\mu_i}(x), \quad \forall x \in \mathcal{X}, \quad i = 0, 1, \dots, \quad (2.88)$$

and the inequality is strict for at least one state. Repeating this procedure only while the current policy is not greedy therefore terminates after finitely many policy improvements at an optimal deterministic stationary policy.

Proof. Policy improvement. For one iteration, greediness and the policy-evaluation equation give

$$T_{\mu_{i+1}} J_{\mu_i} = TJ_{\mu_i} \leq T_{\mu_i} J_{\mu_i} = J_{\mu_i}.$$

Starting from J_{μ_i} , repeated application of the monotone operator $T_{\mu_{i+1}}$ produces a nonincreasing sequence. Contraction makes that sequence converge to its unique fixed point $J_{\mu_{i+1}}$, proving Equation (2.88). Because μ_i is not greedy, at some state $\bar{x}$ the first inequality above is strict. Every subsequent iterate is no larger than the first one, so the limiting value also satisfies $J_{\mu_{i+1}}(\bar{x}) < J_{\mu_i}(\bar{x})$.

Finite termination and optimality. The finite-state, finite-control model has only finitely many deterministic stationary policies. The strict value improvement above prevents a policy from being revisited. For each fixed policy μ , the policy-evaluation equation has a unique solution J_μ . This means only that revisiting the *same* policy would reproduce its previous value function, contradicting the cumulative strict improvement established above. Hence the sequence cannot continue indefinitely. Notice that a policy change caused only by switching between tied minimizing controls would not imply strict improvement; the algorithm instead stops as soon as the current policy is greedy. At termination, the current policy is greedy with respect to its own value and therefore satisfies

$$TJ_{\mu_i} = T_{\mu_i}J_{\mu_i} = J_{\mu_i}.$$

Uniqueness of the Bellman fixed point then gives $J_{\mu_i} = J^*$, and theorem 2.1 shows that μ_i is optimal. $\square$

Algorithm 2.4: Exact discounted PI

1. **Initialization.** Choose a stationary policy μ_0 , defined for every $x \in \mathcal{X}$, and set $i = 0$.
2. **Policy evaluation.** Evaluate μ_i exactly by solving the coupled linear system Equation (2.85) for the full table J_{μ_i} , equivalently by solving $J_{\mu_i}(x) = (T_{\mu_i}J_{\mu_i})(x)$ simultaneously for every $x \in \mathcal{X}$.
3. **Optimality test.** If $\mu_i(x)$ attains the minimum in Equation (2.87) for every $x \in \mathcal{X}$, return μ_i .
4. **Policy improvement.** Otherwise choose, for every $x \in \mathcal{X}$, a minimizing control in Equation (2.87) to define μ_{i+1} ; increment i and repeat.

Q-factor PI.

PI also has an exact Q-factor form. Evaluation of a fixed policy solves

$$Q_{\mu_i}(x, u) = \mathbb{E}_{w|x, u} [g(x, u, w) + \alpha Q_{\mu_i}(f(x, u, w), \mu_i(f(x, u, w)))], \quad \forall x \in \mathcal{X}, \quad u \in \mathcal{U}(x), \quad (2.89)$$

and policy improvement selects

$$\mu_{i+1}(x) \in \arg \min_{u \in \mathcal{U}(x)} Q_{\mu_i}(x, u), \quad \forall x \in \mathcal{X}. \quad (2.90)$$

By definition, $Q_{\mu_i}(x, u) = \mathbb{E}[g(x, u, w) + \alpha J_{\mu_i}(f(x, u, w)) \mid x, u]$ and $J_{\mu_i}(x) = Q_{\mu_i}(x, \mu_i(x))$. Therefore minimizing $Q_{\mu_i}(x, u)$ compares exactly the same candidate costs as the value-form improvement in Equation (2.87). With the same tie-breaking rule, the exact value and Q-factor

forms generate the same next policy; see Bertsekas (2019, Eqs. (4.35)–(4.36)). Q-factor PI stores more entries, but it exposes the control comparison without an explicit transition expectation at control-selection time.

2.7.4 VI and PI Compared

Both methods reach the same exact optimum under the declared finite discounted assumptions, but their outer iterations do different amounts of work. One VI iteration is a full VI sweep, whereas one PI iteration includes an exact policy evaluation—a linear solve or policy-evaluation sweeps continued to convergence—followed by improvement. Counting outer iterations alone can therefore be misleading. Table 2.3 compares both the work performed within an iteration and the resulting convergence property.

Table 2.3: Computational tradeoffs between exact VI and PI and their Q-factor forms.

Form	Principal advantage	Principal limitation
VI (value form)	Simple VI sweeps, an α -contraction rate, and a computable residual certificate	Convergence is asymptotic and may require many full sweeps when α is near one
PI (value form)	Guaranteed termination after finitely many strict policy changes, and often rapid improvement in practice	Each exact evaluation is a global linear solve or policy-evaluation sweeps continued to convergence
Q-factor forms of VI and PI	Greedy control is available from a stored state–control table and a Q-VI backup component can be sampled	The table is larger; exact VI/PI still uses a model and continuous controls still require minimization

What exact tabular computation requires.

Exact tabular VI and PI ordinarily use the one-stage cost and transition probabilities for every feasible state–control pair and update or evaluate the value over every state. They are therefore explicit-model, full-state computations and are most naturally performed offline. Once the resulting table and policy have been constructed, executing the stored policy can be cheap.

A Q-factor representation makes sample-based computation especially natural: one observed or simulated transition gives the single-component target in Equation (2.83). Such a sample avoids analytically evaluating the conditional expectation for that update. It is nevertheless a sample of an exact Q-backup, not the exact expectation itself. Repeated sample-based learning also requires sufficient state–control coverage and appropriate update rules, as Chapter 4 explains. Conversely, exact tabular Q-factor VI or PI still requires the exact conditional expectation for every feasible state–control pair, whether supplied by an explicit model or an equivalent exact oracle.

Q-factor notation enables model-free learning when transitions are collected without querying a model, but the representation is not intrinsically model-free; simulator-generated targets can be sample-based and model-based at the same time. The next section replaces selected value, expectation, minimization, or policy-evaluation computations with tractable approximations.

2.8 Approximate Infinite-Horizon Dynamic Programming

Exact VI and PI are useful reference algorithms because every Bellman backup, policy evaluation, and greedy improvement is defined on the complete finite-state, finite-control reference model. Large or continuous problems usually require at least one of these computations to be replaced by a tractable approximation. Section 2.4 introduced the same issue for a finite horizon. Figure 2.13 uses the same three terms as Figure 2.5: approximate cost-to-go, approximate expectation, and approximate minimization. The only structural difference is that a stationary infinite-horizon method reuses the same cost-to-go approximation at every stage.

The purpose here is structural: to identify which value object or Bellman computation is approximated and how that choice changes policy construction. Chapter 3 develops the representations and fitting tools. Chapter 4 then turns this taxonomy into executable value-space methods by specifying model- or sample-based target generation, policy evaluation and improvement, error propagation, and online execution.

Approximate minimization

$$\min_{u \in \mathcal{U}(x)} \mathbb{E}_{w|x,u} \left[g(x,u,w) + \alpha \tilde{J}(f(x,u,w)) \right]$$

Computation of $\mathbb{E}$:

Replace $\mathbb{E}$ with nominal values
(certainty equivalence)
Adaptive simulation

Computation of $\tilde{J}$:

Problem approximation
Rollout
Approximate PI
Parametric approximation

Figure 2.13: Three approximation sites in a stationary infinite-horizon Bellman decision: feasible-control minimization, conditional expectation, and the cost-to-go used after the successor state. The stationary setting reuses the same approximate cost-to-go at every stage. Adapted from the value-space organization in Bertsekas (2019, Fig. 4.6.1).

The three sites are design choices rather than mutually exclusive algorithms. A method may approximate one site or several sites together, and problem approximation may first change the

ingredients supplied to the Bellman calculation. This section begins with approximate VI and approximate PI. The four subsections introduce approximate VI, partial policy evaluation through optimistic PI, approximate PI, and then the remaining approximation sites together with rollout.

2.8.1 Approximate VI

Approximate VI retains the optimality-backup structure of exact VI but replaces an exact value table by a fitted function. Given a current approximation $\tilde{J}_i$, choose a set of training states $\mathcal{S}_i \subseteq \mathcal{X}$, form Bellman targets there, and fit the next approximation:

$$y_i(x) := (T\tilde{J}_i)(x), \quad \tilde{J}_{i+1}(x) \approx y_i(x), \quad x \in \mathcal{S}_i. \quad (2.91)$$

The approximation is then used at states outside $\mathcal{S}_i$; its generalization there is not asserted by the fitting relation in Equation (2.91). Here $\approx$ denotes fitting under a declared loss, not a pointwise guarantee. If $\mathcal{S}_i = \mathcal{X}$ and the targets are stored exactly, the update reduces to exact VI. The same backup–fit idea applies to Q-factors. The representation and fitting method are developed in Chapter 3; Chapter 4 distinguishes exact-model, sampled-model, and physical-transition targets.

This apparently small replacement changes the convergence question. A useful discounted-model analysis assumes a uniform backup error

$$\|\tilde{J}_{i+1} - T\tilde{J}_i\|_\infty \leq \delta, \quad i = 0, 1, \dots \quad (2.92)$$

Under this condition, one can bound the asymptotic value and greedy-policy errors. Natural fitted or projected VI, however, need not satisfy a known, sufficiently small uniform bound of this form over all iterations: fitting error can be amplified because the composition of a Bellman update and a projection need not be a contraction (Bertsekas 2019, Eq. (4.22) and Example 4.4.1). This is an important disadvantage relative to the evaluation–improvement organization of approximate PI, where evaluation and improvement errors can be diagnosed separately. It does not imply that approximate PI is automatically stable or monotonically improving. Chapter 4 treats both error mechanisms in detail.

2.8.2 Partial Policy Evaluation and Optimistic PI

Exact PI solves a coupled policy-evaluation equation before every policy improvement. That global solve can dominate the computation even when an intermediate evaluation would already contain enough information to select the next greedy policy. Optimistic PI replaces exact policy evaluation by a finite number of fixed-policy VI sweeps. Starting from a value table J_i , it first computes a greedy policy

$$\mu_i(x) \in \arg \min_{u \in \mathcal{U}(x)} \mathbb{E}_{w|x,u} [g(x, u, w) + \alpha J_i(f(x, u, w))], \quad \forall x \in \mathcal{X}, \quad (2.93)$$

and then performs $m_i \geq 1$ evaluation sweeps:

$$\begin{aligned} \hat{J}_{i,0} &= J_i, \\ \hat{J}_{i,m+1}(x) &= (T_{\mu_i} \hat{J}_{i,m})(x), \quad \forall x \in \mathcal{X}, \quad m = 0, \dots, m_i - 1, \\ J_{i+1} &= \hat{J}_{i,m_i}. \end{aligned} \quad (2.94)$$

Equivalently,

$$J_{i+1} = T_{\mu_i}^{m_i} J_i, \quad T_{\mu_i} J_i = T J_i. \quad (2.95)$$

The equality on the right follows from the greedy choice in Equation (2.93). When $m_i = 1$, the method is exact VI. If each fixed-policy iteration is continued to its fixed point, the method becomes exact PI. Intermediate depths trade a cheaper policy evaluation for more frequent policy improvement. Related recursions are also called *modified PI*; this book uses *optimistic PI* for the precisely defined recursion above, following Bertsekas (2019, Eqs. (4.33)–(4.34)). Here “optimistic” does not refer to exploration bonuses or uncertainty optimism.

Algorithm 2.5: Optimistic PI

1. **Initialization.** Choose a finite-valued table J_0 , choose a positive integer m_i for each outer iteration, and set $i = 0$.
2. **Policy improvement.** For every $x \in \mathcal{X}$, choose $\mu_i(x)$ by the full Bellman minimization in Equation (2.93).
3. **Partial policy evaluation.** Set $\hat{J}_{i,0} = J_i$ and apply the fixed-policy Bellman operator T_{μ_i} exactly m_i times according to Equation (2.94).
4. **Outer update.** Set $J_{i+1} = \hat{J}_{i,m_i}$, increment i , and repeat until a declared Bellman-residual tolerance is met. An unchanged greedy policy at one outer iteration is not, by itself, a stopping certificate for optimistic PI.

Proposition 2.6 (Convergence of optimistic PI). *Under assumption 2.3, let J_0 be any value table, let every greedy minimization and policy-evaluation sweep in Equations (2.93) and (2.94) be exact, and let every m_i be a finite positive integer. Then*

$$\lim_{i \rightarrow \infty} \|J_i - J^*\|_\infty = 0, \quad \lim_{i \rightarrow \infty} \|J_{\mu_i} - J^*\|_\infty = 0. \quad (2.96)$$

This is the discounted finite-state, finite-control result of Bertsekas (2019, Prop. 4.5.2). It does not assert monotonic policy improvement or finite termination, and it does not automatically extend to regression, sampling, or approximate minimization errors. Those practical variants belong to approximate PI and are developed in Chapter 4.

2.8.3 Approximate PI

When a state space is too large for exact PI, the policy-evaluation step, the policy-improvement step, or both may be approximate. Approximate PI therefore retains a visible evaluation–improvement loop:

$$\tilde{J}_{\mu_i} \approx J_{\mu_i} \quad \rightarrow \quad \mu_{i+1} \approx \text{Greedy}(\tilde{J}_{\mu_i}). \quad (2.97)$$

The evaluated object may instead be $\tilde{Q}_{\mu_i}$. Policy evaluation may use an analytical model, simulated trajectories, or physical-system data. Policy improvement may use an exact minimization,

an approximate optimizer, or sampled improved controls. Finally, the improved policy may be stored directly on a finite state set or fitted by an approximation in policy space.

With exact minimization and an approximate value, the improvement step is

$$\mu_{i+1}(x) \in \arg \min_{u \in \mathcal{U}(x)} \mathbb{E}_{w|x,u} [g(x, u, w) + \alpha \tilde{J}_{\mu_i}(f(x, u, w))], \quad \forall x \in \mathcal{X}, \quad (2.98)$$

or, with an approximate Q-factor, $\mu_{i+1}(x) \in \arg \min_{u \in \mathcal{U}(x)} \tilde{Q}_{\mu_i}(x, u)$ for every $x \in \mathcal{X}$. Approximation can therefore enter evaluation, improvement, and policy representation as separate design choices, as shown in Figure 2.14. Unlike exact PI, approximate PI need not improve monotonically: evaluation error, incomplete state coverage, and improvement error can cause degradation or policy oscillation. Chapter 4 states the corresponding error assumptions and performance region; the present section establishes only the architecture.

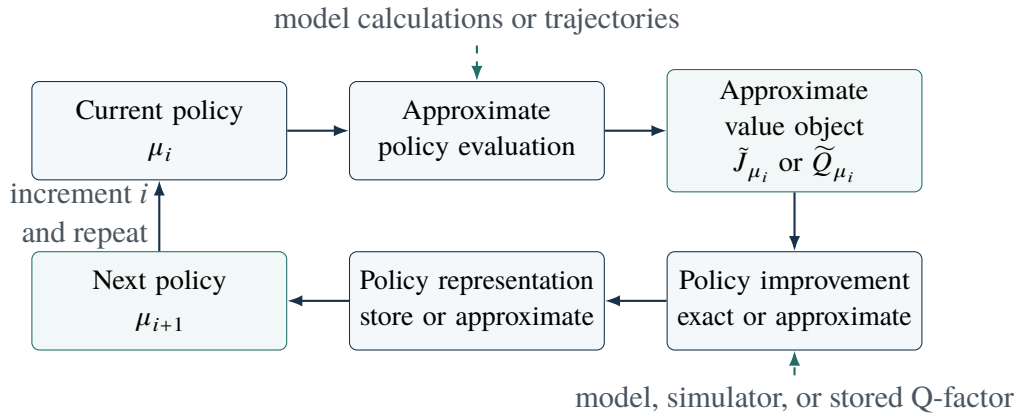


Figure 2.14: The approximate PI loop. Policy evaluation and policy improvement may be exact or approximate in different combinations. A model-based variant uses transition and cost models in one or both blocks; a simulation-based variant estimates the required quantities from trajectories. The improved policy may be stored directly or represented by a fitted policy.

2.8.4 Other Approximation Sites and Rollout

The remaining approximation routes mirror the finite-horizon organization in Section 2.4.

Problem approximation.

The dynamics, costs, constraints, uncertainty description, or horizon may be replaced by a tractable surrogate before a Bellman method is applied. The link-domain HEV construction in Section 2.5.2 is one example.

Approximate expectation.

A conditional expectation may use certainty equivalence, scenarios, Monte Carlo samples, or adaptive simulation. The source of the samples remains a separate model-use question: simulator

samples are sample-based and model-based, whereas physical transitions need not query an explicit model.

Approximate minimization.

The feasible-control search may be restricted to selected candidates, solved locally, or stopped early. A fitted policy may supply candidate controls or an initialization. Feasibility and the availability of a fallback control must be preserved explicitly.

Pure rollout as one policy improvement.

Let μ_{base} be a stationary base policy and let $J_{\mu_{\text{base}}}$ be its exact cost-to-go function. Pure rollout evaluates that base policy and then performs one greedy improvement. If a restricted candidate set $\overline{\mathcal{U}}(x)$ is used, it must retain the base action:

$$\mu_{\text{base}}(x) \in \overline{\mathcal{U}}(x) \subseteq \mathcal{U}(x), \quad \forall x \in \mathcal{X}. \quad (2.99)$$

The rollout policy is

$$\mu_{\text{ro}}(x) \in \arg \min_{u \in \overline{\mathcal{U}}(x)} \mathbb{E}_{w|x,u} [g(x, u, w) + \alpha J_{\mu_{\text{base}}}(f(x, u, w))], \quad \forall x \in \mathcal{X}. \quad (2.100)$$

The base action in Equation (2.99) gives the minimization a feasible benchmark at every state.

Proposition 2.7 (Cost improvement by pure rollout). *Under assumption 2.3 and Equation (2.99), suppose the base-policy value, conditional expectation, and minimization in Equation (2.100) are exact. Then*

$$J_{\mu_{\text{ro}}}(x) \leq J_{\mu_{\text{base}}}(x), \quad \forall x \in \mathcal{X}. \quad (2.101)$$

Proof. Because the base action remains feasible, $T_{\mu_{\text{ro}}} J_{\mu_{\text{base}}} \leq T_{\mu_{\text{base}}} J_{\mu_{\text{base}}} = J_{\mu_{\text{base}}}$. Repeated application of the monotone operator $T_{\mu_{\text{ro}}}$ preserves this inequality, and its discounted contraction converges to $J_{\mu_{\text{ro}}}$. Hence $J_{\mu_{\text{ro}}} \leq J_{\mu_{\text{base}}}$ componentwise. $\square$

When $\overline{\mathcal{U}}(x) = \mathcal{U}(x)$ at every state, pure rollout is one exact PI improvement. With a proper restricted candidate set, it is instead an exact restricted policy-improvement step; retaining the base action is what preserves the cost-improvement conclusion. This is the rollout result of Bertsekas (2019, Prop. 4.6.2). The policy evaluation that estimates $J_{\mu_{\text{base}}}$ may use equations or Monte Carlo simulation; stochastic simulation generally requires averaging multiple trajectories. For an infinite discounted trajectory, the simulated return is truncated in practice. Once a finite sample estimate or an approximate terminal value replaces exact $J_{\mu_{\text{base}}}$, the unconditional improvement guarantee above no longer follows.

Truncated rollout and multistep lookahead.

Pure rollout uses the exact infinite-horizon value of the base policy. In practice, that tail is often replaced by $m \geq 1$ simulated or predicted base-policy stages followed by a designed or learned terminal approximation $\hat{J}_T : \mathcal{X} \rightarrow \mathbb{R}$. The resulting truncated-rollout tail evaluator is

$$\hat{J}_{\mu_{\text{base}},m}(x) = \mathbb{E}^{\mu_{\text{base}}} \left[\sum_{t=0}^{m-1} \alpha^t g(x_t, \mu_{\text{base}}(x_t), w_t) + \alpha^m \hat{J}_T(x_m) \mid x_0 = x \right], \quad \forall x \in \mathcal{X}. \quad (2.102)$$

One-step truncated rollout attaches this evaluator after the candidate root control. More generally, an ℓ -step lookahead chooses feedback decisions $v_0, \dots, v_{\ell-1}$ to minimize

$$\mathbb{E} \left[\sum_{t=0}^{\ell-1} \alpha^t g(x_t, v_t(x_t), w_t) + \alpha^\ell \hat{J}_{\mu_{\text{base}},m}(x_\ell) \mid x_0 = x \right], \quad (2.103)$$

and executes only the first optimized control $v_0(x)$. In a stochastic problem, the v_t are state-contingent decisions on the lookahead tree, not one open-loop input sequence. Figure 2.15 separates the lookahead depth ℓ , base-policy rollout depth m , and terminal approximation.

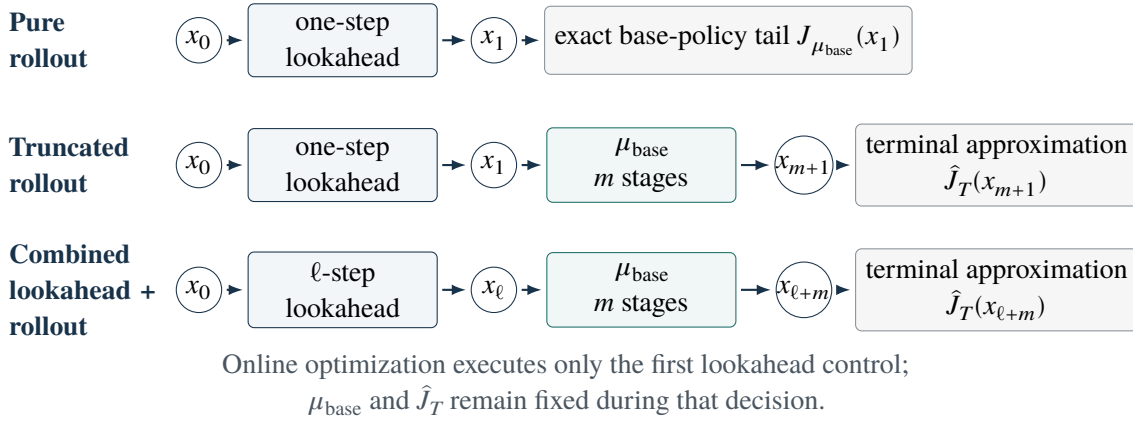


Figure 2.15: Independent comparison of pure rollout, truncated rollout, and combined multistep lookahead. In a stochastic problem, each lookahead block represents state-contingent decisions on a scenario tree. Only the first optimized control is applied before the calculation is repeated. Conceptually informed by Bertsekas (2019, Sec. 4.6.2 and Fig. 4.6.3).

Pure rollout therefore guarantees policy improvement under the exactness and base-action conditions above, and it coincides with one exact PI improvement when $\overline{\mathcal{U}}(x) = \mathcal{U}(x)$ at every state. Truncated rollout and combined multistep lookahead replace the exact base-policy value by finite computation and a terminal approximation. Chapter 4 develops their performance bounds, sampled variants, online computation, and relationship to MPC.

2.9 General Finite-to-Infinite-Horizon Reformulation

Every finite-horizon Markov problem of the form developed in Sections 2.2 and 2.3 can be represented as a stationary infinite-horizon total-cost problem. The construction attaches the stage to the state and adds a cost-free absorbing state. It is general for the finite-horizon Markov class declared in this chapter; partial observation or a state that omits other predictive information still requires an appropriate information state.

The explicit stage tag is needed when the same physical state can occur at different stages but the remaining horizon, dynamics, cost, constraints, or disturbance law differ. It is redundant if the original state already contains the clock or phase, or if membership in the stage spaces uniquely identifies the stage. For state spaces $\mathcal{X}_0, \dots, \mathcal{X}_N$, define the disjoint augmented space

$$\mathcal{Z} = \left(\bigcup_{k=0}^N \{k\} \times \mathcal{X}_k \right) \cup \{\Delta\}, \quad z = (k, x), \quad (2.104)$$

where Δ is absorbing. To define one stationary controlled transition law, the stage tag selects the original disturbance space and law: when $z = (k, x)$ with $k < N$, and u is applied, $w \in \mathcal{W}_k$ is drawn from $P_k^w(\cdot | x, u)$. For $k < N$, define

$$\begin{aligned} \bar{f}((k, x), u, w) &= (k + 1, f_k(x, u, w)), & \bar{g}((k, x), u, w) &= g_k(x, u, w), \\ \bar{\mathcal{U}}((k, x)) &= \mathcal{U}_k(x). \end{aligned} \quad (2.105)$$

Thus all stage dependence is carried by z ; the augmented transition, cost, feasible-control set, and disturbance rule do not depend on external time. A deterministic model is covered by taking each conditional disturbance law to be degenerate. At (N, x) , introduce a dummy control u_Δ , charge the terminal cost, and move to Δ :

$$\begin{aligned} \bar{\mathcal{U}}((N, x)) &= \bar{\mathcal{U}}(\Delta) = \{u_\Delta\}, \\ \bar{f}((N, x), u_\Delta, w) &= \Delta, & \bar{g}((N, x), u_\Delta, w) &= g_N(x), \\ \bar{f}(\Delta, u_\Delta, w) &= \Delta, & \bar{g}(\Delta, u_\Delta, w) &= 0. \end{aligned} \quad (2.106)$$

The disturbance law at the terminal layer and at Δ may be any fixed degenerate law because it affects neither transition nor cost. The original problem has N genuine control decisions; the transition from (N, x_N) to Δ is only a bookkeeping step that records $g_N(x_N)$.

The corresponding infinite-horizon criterion is the undiscounted total cost

$$\bar{J}_{\bar{\mu}}(z) = \mathbb{E}^{\bar{\mu}} \left[\sum_{t=0}^{\infty} \bar{g}(z_t, \bar{\mu}(z_t), w_t) \mid z_0 = z \right]. \quad (2.107)$$

Every trajectory reaches Δ after the prescribed finite number of layers, so the sum contains exactly the original running and terminal costs. In the finite-state specialization, this is an SSP whose transient portion is acyclic; Δ supplies the only self-loop. If the original finite-horizon objective is discounted, its stage weights must be retained in g_k and g_N ; adding a discount factor that was not present changes the problem.

Figure 2.16 summarizes the construction, following the general reformulation in Bertsekas (2025, Sec. 1.6.4) and its finite-state SSP form in Bertsekas (2019, Sec. 4.2).

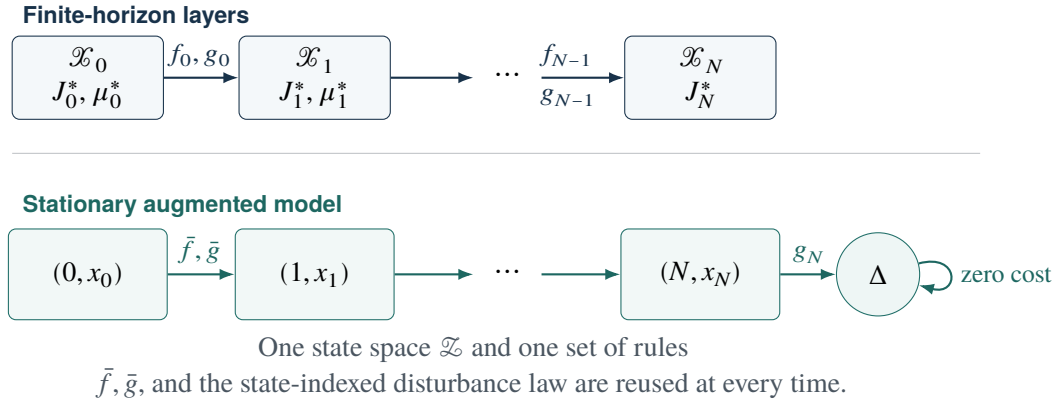


Figure 2.16: A finite-horizon Markov problem represented on a stationary augmented state space. The stage tag advances from $(0, x_0)$ through (N, x_N) before the process enters the cost-free absorbing state Δ .

Let $\bar{J}^*$ denote the augmented optimal value. For any selected optimal sequence μ^* of the original problem, define the corresponding stationary augmented policy by using the same control on each tagged decision layer and the dummy control thereafter. Then

$$\begin{aligned} \bar{J}^*((k, x)) &= J_k^*(x), & k &= 0, \dots, N, \\ \bar{\mu}^*((k, x)) &= \mu_k^*(x), & k &= 0, \dots, N-1. \end{aligned} \quad (2.108)$$

At layer N and at Δ , the only control is u_Δ . Hence a stationary rule on $\mathcal{Z}$ retains, rather than erases, the original stage dependence: the stage is now an argument of the rule.

This is the standard time-augmentation construction described in Bertsekas (2025, Sec. 1.6.4) and Bertsekas (2019, Sec. 4.2). The mapping $\bar{\mu}((k, x)) = \mu_k(x)$ is one-to-one on the decision layers, with the dummy control fixed at layer N and at Δ . Equations (2.105) and (2.106) preserve the admissible controls, conditional disturbance laws, physical state trajectories, running costs, and terminal cost. Corresponding policies therefore have the same expected total cost, which gives Equation (2.108). This is a cost- and law-preserving representation, not a general SSP convergence theorem.

Applying VI and PI to the augmented SSP.

For this algorithmic paragraph, restrict attention to the finite-state, finite-control specialization with real-valued costs. The augmented model is an undiscounted SSP, so its Bellman equations have the same algebraic form as the discounted equations of Section 2.7 with the external factor α omitted. It should not, however, be described as a discounted problem with $\alpha = 1$: the convergence argument used here comes from finite termination and the acyclic stage layers,

not from discounted contraction. For any finite-valued table $\bar{J}$ satisfying the terminal boundary $\bar{J}(\Delta) = 0$, define

$$(\bar{T}\bar{J})(z) = \min_{u \in \mathcal{U}(z)} \mathbb{E}_{w|z,u} [\bar{g}(z, u, w) + \bar{J}(\bar{f}(z, u, w))], \quad \forall z \in \mathcal{Z} \setminus \{\Delta\}, \quad (2.109)$$

$$(\bar{T}_{\bar{\mu}}\bar{J})(z) = \mathbb{E}_{w|z, \bar{\mu}(z)} [\bar{g}(z, \bar{\mu}(z), w) + \bar{J}(\bar{f}(z, \bar{\mu}(z), w))], \quad \forall z \in \mathcal{Z} \setminus \{\Delta\}. \quad (2.110)$$

Here $\bar{T}$ is the Bellman optimality operator and $\bar{T}_{\bar{\mu}}$ is the fixed-policy operator for the augmented model; when $z = (k, x)$, the notation $\mathbb{E}_{w|z,u}$ denotes expectation under the original stage- k law for $k < N$ and under the fixed degenerate law at $k = N$. Exact VI is therefore

$$\bar{J}_{i+1}(z) = (\bar{T}\bar{J}_i)(z), \quad \forall z \in \mathcal{Z} \setminus \{\Delta\}, \quad \bar{J}_i(\Delta) = 0, \quad i = 0, 1, \dots \quad (2.111)$$

The first synchronous sweep makes layer N exact because it depends only on the fixed boundary $\bar{J}(\Delta) = 0$. Each subsequent sweep propagates exactness backward by one layer, so all layers are exact after at most $N + 1$ sweeps. Replacing the minimization by a fixed policy gives the same finite propagation for policy evaluation. Exact PI alternates the following two operations:

Policy evaluation.

$$\begin{aligned} \bar{J}_{\bar{\mu}_i}(z) &= (\bar{T}_{\bar{\mu}_i}\bar{J}_{\bar{\mu}_i})(z), \quad z \in \mathcal{Z} \setminus \{\Delta\}, \\ \bar{J}_{\bar{\mu}_i}(\Delta) &= 0. \end{aligned} \quad (2.112)$$

Policy improvement.

For every decision-layer state $z = (k, x)$ with $k < N$, set

$$\bar{\mu}_{i+1}(z) \in \arg \min_{u \in \mathcal{U}(z)} \mathbb{E}_{w|z,u} [\bar{g}(z, u, w) + \bar{J}_{\bar{\mu}_i}(\bar{f}(z, u, w))]. \quad (2.113)$$

The dummy control remains fixed at layer N and at Δ . Thus no new VI or PI update is required after reformulation: the state changes from x to $z = (k, x)$ and the terminal boundary $\bar{J}(\Delta) = 0$ is held fixed.

The reformulation does not make the exact finite-horizon computation smaller. A finite-state model contains $1 + \sum_{k=0}^N |\mathcal{X}_k|$ augmented states, and backward DP remains the most direct exact solver for its acyclic layers. PI may require fewer outer improvement checks than synchronous VI requires sweeps, but each PI check also contains a complete policy evaluation; outer-iteration counts therefore do not establish a runtime advantage. The benefit of reformulation is structural: the sequence $\{J_k^*, \mu_k^*\}$ becomes one stationary value and policy, so total-cost or episodic variants of VI, PI, value approximation, rollout, and sample-based learning can share one Bellman interface. It also makes explicit which clock, phase, or remaining-horizon information an RL state must contain. These are advantages of conceptual and algorithmic transfer, not reductions in the number of state–stage pairs.

2.10 Mobility Case: Finite and Periodic Slalom Tracking

This case study answers two different questions with one declared finite-state model. First, does a nonstationary finite-horizon problem give exactly the same value and policy after the stage-augmented SSP reformulation of Section 2.9? Second, on a continuing task, how does the control law obtained from a finite preview approach the genuine infinite-horizon optimum as that preview grows? Keeping these questions separate is essential: the first comparison changes only the representation, whereas the second changes the control objective.

An auditable teaching model.

A high-fidelity slalom model would combine path-error kinematics with the vehicle dynamics introduced in Chapter 1. The present example is not obtained by discretizing that full model. Instead, it uses the normalized two-state abstraction in Figure 2.17. Let r_k be the prescribed lateral reference, y_k the normalized lateral position, and e_k the normalized tracking-error coordinate. Before grid saturation, $e_k = y_k - r_k$, and the auxiliary increment η_k gives $y_{k+1} = y_k + \eta_k$. The tabular update below clips e_k ; at an active boundary it is a saturated surrogate rather than an independently propagated identity. When y_k is plotted later, it is reconstructed as $e_k + r_k$. The input u_k is a normalized steering correction and w_k is an additive disturbance to the lateral-motion increment. Thus

$$e_{k+1} = \text{clip}(e_k + \eta_k - \Delta r_k, -6, 6), \quad \Delta r_k := r_{k+1} - r_k, \quad (2.114)$$

$$\eta_{k+1} = \text{clip}(\eta_k + u_k + w_k, -2, 2). \quad (2.115)$$

The finite grids and disturbance law are

$$e_k \in \{-6, -5, \dots, 6\}, \quad \eta_k \in \{-2, -1, \dots, 2\}, \quad u_k, w_k \in \{-1, 0, 1\}, \quad (2.116)$$

with probabilities 0.10, 0.80, and 0.10 for $w_k = -1, 0, 1$, respectively. Here $\text{clip}(q, \underline{q}, \bar{q}) := \min\{\bar{q}, \max\{q, \underline{q}\}\}$ keeps a successor on the declared grid. The stage and terminal costs are

$$g(e, \eta, u) = e^2 + 0.25\eta^2 + 0.50u^2, \quad (2.117)$$

$$g_N(e, \eta) = 4e^2 + \eta^2. \quad (2.118)$$

For the finite study, set $N = 12$ and read the known reference sequence as

$$(r_0, \dots, r_{12}) = (0, 1, 2, 3, 2, 1, 0, -1, -2, -3, -2, -1, 0). \quad (2.119)$$

The physical state has $13 \times 5 = 65$ grid points and there are three controls. The model is intentionally modest; useful steering must anticipate changes in Δr_k . All computations below use the known transition probabilities at every declared state–control pair. They are therefore exact, explicit-model Bellman calculations for this grid, even though the grid itself approximates vehicle motion.

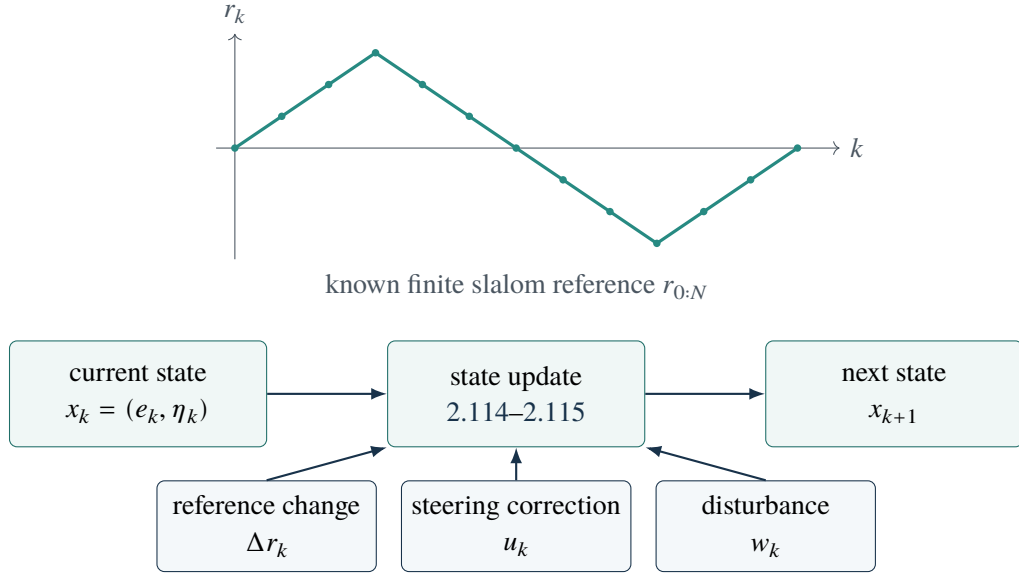


Figure 2.17: Declared slalom-tracking teaching model. The prescribed reference changes with stage, while the finite state records tracking error and a normalized lateral-motion increment. Steering and disturbance change the successor state. The model isolates the horizon and state-augmentation ideas; it is not a calibrated vehicle model.

2.10.1 Exact Equivalence of Finite DP and an Augmented SSP

For the prescribed finite slalom in Equation (2.119), consider the undiscounted objective

$$J_\mu(e_0, \eta_0) = \mathbb{E}^\mu \left[\sum_{k=0}^{N-1} g(e_k, \eta_k, u_k) + g_N(e_N, \eta_N) \mid e_0, \eta_0 \right]. \quad (2.120)$$

Because $N = 12$ is fixed in this subsection, no extra horizon superscript is needed: $J_k^*(e, \eta)$ and $\mu_k^*(e, \eta)$ denote the optimal cost-to-go and feedback law at stage k . Backward DP computes these stage-dependent tables directly.

The known reference does not make the model stationary in the physical state (e, η) . It makes the future increments Δr_k predictable, but the successor associated with the same (e, η, u, w) still changes with k . Applying Section 2.9 gives the augmented state $z = (k, e, \eta)$ and the single transition rule

$$\tilde{f}((k, e, \eta), u, w) = (k + 1, \text{clip}(e + \eta - \Delta r_k, -6, 6), \text{clip}(\eta + u + w, -2, 2)), \quad k < N. \quad (2.121)$$

The terminal transition charges Equation (2.118) before entering Δ . Equation (2.121) is stationary on the augmented state space: k is now a state coordinate rather than an external clock that changes the rule.

The augmented VI and PI computations use Equations (2.111) to (2.113) without importing a discounted-contraction argument. By the layered argument in Section 2.9, synchronous VI propagates the terminal cost backward through one tagged layer per sweep. PI evaluates each

augmented policy on the same acyclic layers and then improves all decision-layer controls. For reproducibility, VI starts from the zero table, PI starts from $u = 0$, and ties are resolved by the fixed control enumeration $-1, 0, 1$.

Table 2.4 records the numerical audit. The 13 VI sweeps correspond to the 13 tagged state layers, whereas each of the six PI improvement checks includes a complete policy evaluation. Both methods recover the backward-DP value and policy at every declared state–stage pair; the iteration counts are not runtime comparisons.

Table 2.4: Exact numerical verification for the $N = 12$ finite slalom.

Computation	Convergence record	Agreement with backward DP
Backward DP	One reverse pass through the 12 decision layers	$J_0^*(0, 0) = 27.3538$
Augmented-SSP VI	Exact after 13 productive sweeps; final Bellman residual 0	Value sup difference 0; policy mismatches 0
Augmented-SSP PI	Control changes [661, 140, 58, 14, 1, 0] over six improvement checks	Value sup difference 0; policy mismatches 0

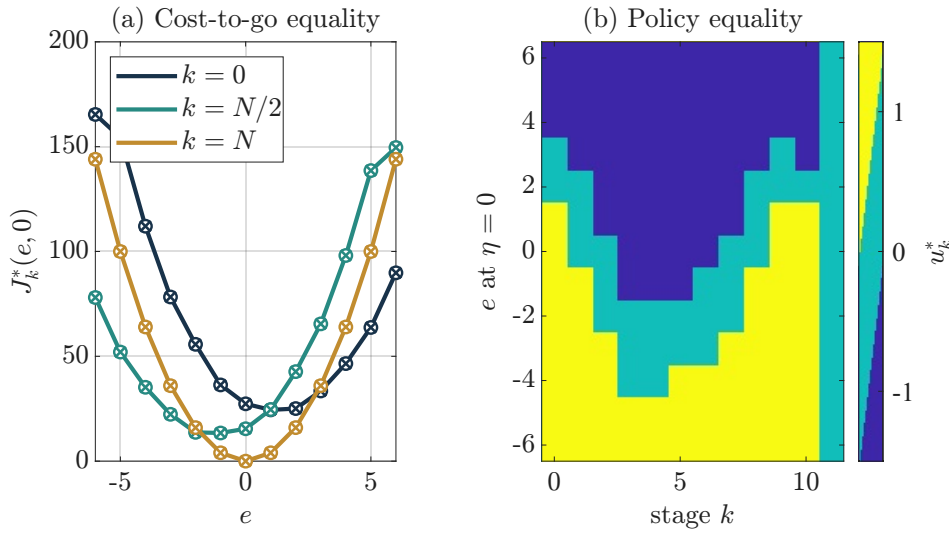


Figure 2.18: Equality of finite-horizon backward DP and exact VI/PI on the augmented SSP. Left: solid curves, open circles, and crosses denote backward DP, VI, and PI; coincident markers show cost-to-go equality on the displayed $\eta = 0$ slices. Right: their common policy map. The numerical audit covers every state–stage pair.

Figure 2.18 checks more than one initial condition. The method markers coincide at every plotted state on the slice $\eta = 0$, and the right panel shows their common stage-dependent control

map on the same slice. The exported audit tables additionally compare all $12 \times 65 = 780$ decision-layer states, for which VI and PI both have zero policy mismatches.

Figure 2.19 separates solver convergence from the equality display. The upper panel shows the sup-norm error from the complete backward-DP value table; it vanishes after the thirteenth productive sweep, when information has propagated through all 13 tagged layers. The lower panel shows how many state–stage controls change at each PI check and the value error after its exact policy evaluation. The smaller number of PI checks does not show that PI is faster: one PI check contains a policy-evaluation computation, whereas backward DP is already a direct reverse pass through an acyclic graph.

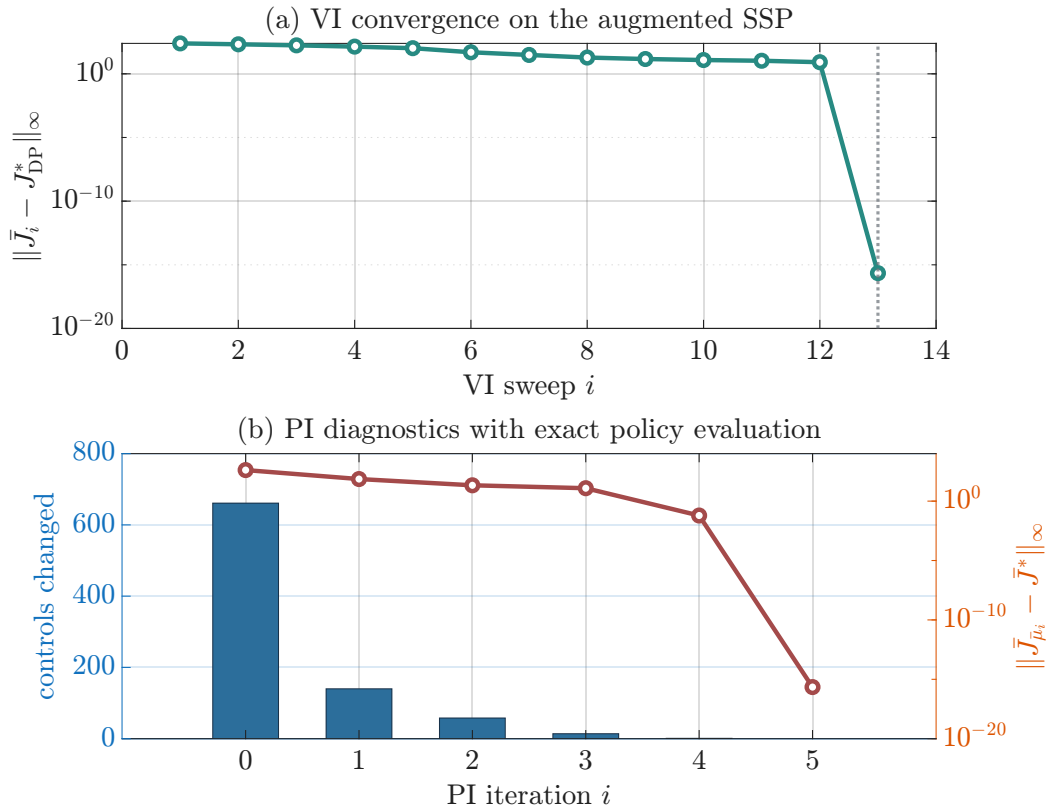


Figure 2.19: Separate convergence diagnostics for augmented-SSP VI and PI. The VI curve measures error from the backward-DP value table; each PI bar follows a complete policy evaluation.

2.10.2 Finite Preview Versus the Periodic Infinite-Horizon Optimum

The preceding equivalence preserves one finite episode. It does not answer what happens when slalom tracking continues. To pose that different question, repeat the same 12-stage reference

cycle indefinitely:

$$r_{k+L} = r_k, \quad L = 12. \quad (2.122)$$

To identify which transition comes next, let $\phi_k = k \bmod L \in \{0, \dots, L-1\}$ and define $\Delta r_{\phi_k} = r_{\phi_{k+1}} - r_{\phi_k}$ with $r_L = r_0$. This phase cannot be replaced by the current reference value alone: the same value occurs on both rising and falling portions of the cycle but has different successors. With $\zeta_k = (\phi_k, e_k, \eta_k)$, the model has one phase-augmented state space and one stationary transition rule. The reference is predictable because the cycle is known; the process is stationary because the phase needed to select the next increment is included in the state.

The continuing benchmark minimizes

$$J_\mu(\zeta) = \mathbb{E}^\mu \left[\sum_{k=0}^{\infty} \alpha^k g(e_k, \eta_k, u_k) \mid \zeta_0 = \zeta \right], \quad \alpha = 0.97. \quad (2.123)$$

This is a stationary discounted problem, not another representation of Equation (2.120): the terminal cost has been removed and the slalom cycle continues forever. Exact VI and PI on the $12 \times 65 = 780$ phase–state pairs produce the benchmark J^* and μ^* . Write ζ^+ for the random successor obtained from Equations (2.114), (2.115) and (2.122).

For comparison, consider the following H -stage finite-horizon OCP at a current phase-augmented state ζ . Its local stage index is reset to $j = 0$, and its terminal cost is zero. Let $\nu = (\nu_0, \dots, \nu_{H-1})$ denote an admissible sequence of feedback laws, with $u_j = \nu_j(\zeta_j) \in \mathcal{U}(\zeta_j)$. Subject to the phase-augmented dynamics defined above, the problem is

$$J_{0|H}^*(\zeta) := \min_{\nu} \mathbb{E}^\nu \left[\sum_{j=0}^{H-1} \alpha^j g(e_j, \eta_j, u_j) \mid \zeta_0 = \zeta \right]. \quad (2.124)$$

Algorithm 2.2 solves this OCP by backward DP, with zero terminal cost, and returns the stage-dependent feedback laws $\mu_{0|H}^*, \dots, \mu_{H-1|H}^*$. The resulting H -stage finite-preview policy is the stationary feedback law obtained by retaining only the first law from this truncated problem:

$$u_k = \mu_H(\zeta_k), \quad \mu_H(\zeta) := \mu_{0|H}^*(\zeta), \quad \forall \zeta. \quad (2.125)$$

At the next physical stage, the local index is reset and the same H -stage OCP is posed from the newly observed state. Thus the remaining planning-tail laws $\mu_{1|H}^*, \dots, \mu_{H-1|H}^*$ are not successively deployed. In this finite-state example, the first-action map μ_H is tabulated offline for every state so that it can be audited; an online implementation could instead solve the current-state OCP when the state is observed. This repeated first-action implementation is the finite-state DP counterpart of receding-horizon MPC.

Figure 2.20 fixes the comparison. The exact continuing policy and the two finite-preview policies μ_3 and μ_8 are all evaluated under Equation (2.123) from the same initial state. Thus only the preview length varies between the two controllers, so the comparison does not mix controller structure with horizon length.

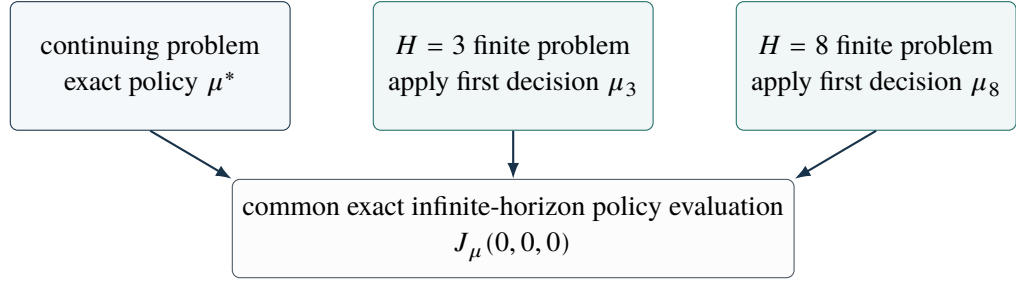


Figure 2.20: Comparison used for the continuing slalom. The infinite-horizon policy is the exact benchmark; the two finite-preview policies differ only in planning horizon. All achieved costs are obtained from the same exact infinite-horizon policy-evaluation equation.

The exact periodic benchmark is independently certified by VI and PI. VI reaches a Bellman residual of 9.49×10^{-12} after 861 sweeps. Across seven improvement checks, PI changes 646, 206, 106, 40, 18, 2, and finally 0 controls. Their values agree to numerical tolerance (sup-norm difference 2.55×10^{-10}), and their policies agree at all 780 phase–state pairs. These iteration counts do not compare runtime because exact PI evaluation and one VI sweep have different computational costs.

Table 2.5: Finite-preview policies evaluated under the common periodic infinite-horizon objective from $(\phi, e, \eta) = (0, 0, 0)$. A mismatch is a phase–state pair at which the finite-preview action differs from μ^* .

Controller	H	Achieved cost	Cost gap	Policy mismatches
Infinite-horizon optimum	—	60.6497	0	0 of 780
Finite-preview policy	3	115.7680	55.1183	134 of 780 (17.18%)
Finite-preview policy	4	65.6309	4.9812	48 of 780 (6.15%)
Finite-preview policy	8	60.9313	0.2817	16 of 780 (2.05%)
Finite-preview policy	12	60.6557	0.0060	2 of 780 (0.26%)
Finite-preview policy	16	60.6497	0	0 of 780

Table 2.5 reports the *achieved* cost of each finite-preview policy, obtained by exact infinite-horizon policy evaluation; $J_{0|H}^*$ is the optimal cost-to-go of the finite-horizon OCP used to construct μ_H ; it is not the infinite-horizon cost J_{μ_H} achieved by repeatedly applying that law. The short preview $H = 3$ misses 134 of the benchmark actions and has a large cost gap. At $H = 8$, only 16 actions differ and the exact cost gap is 0.2817. The first horizon whose policy matches μ^* at every phase–state pair is $H = 16$. The decreasing gaps shown here are numerical findings for the declared model, not a general theorem that every induced finite-horizon policy improves monotonically with H .

Figure 2.21 extends the selected rows of Table 2.5 to $H = 1, \dots, 30$. Every quantity in the figure now evaluates the deployed finite-preview policy μ_H : the upper panel reports its achieved

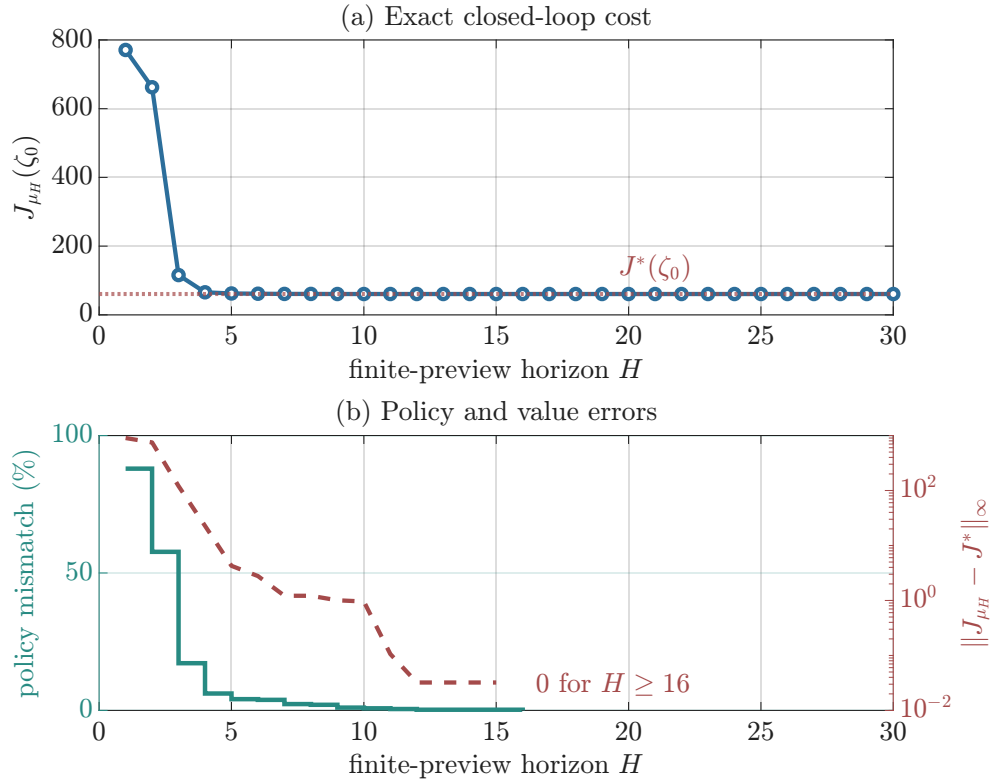


Figure 2.21: Effect of finite preview on the periodically continuing slalom. The upper panel uses exact infinite-horizon policy evaluation for every finite-preview policy. The lower panel distinguishes policy mismatch from the sup-norm error of that policy's infinite-horizon value.

cost from ζ_0 , while the lower panel reports the action-mismatch rate and the all-state policy-value error $\|J_{\mu_H} - J^*\|_\infty$. The planning value $J_{0|H}^* = T^H 0$ is a different object. Discounted contraction gives $J_{0|H}^* \rightarrow J^*$ as $H \rightarrow \infty$, so its truncation error is mathematically meaningful, but it is not the achieved value of μ_H and is therefore not plotted here.

Figure 2.22 separates one realized closed-loop behavior from the exact performance evaluation. The upper panel and lower-left panel use one common 60-stage disturbance sequence to make the different tracking and control actions visible. The lower-right panel does not use that single realization. For a fixed initial state, define the expected partial cost

$$C_K^\mu := \mathbb{E}^\mu \left[\sum_{k=0}^{K-1} \alpha^k g(e_k, \eta_k, u_k) \mid \zeta_0 = (0, 0, 0) \right].$$

The panel propagates the state distribution exactly and plots C_K^μ . As K grows, the curves approach the exact infinite sums reported in Table 2.5, preserving the expected-cost ordering $J^* < J_{\mu_8} < J_{\mu_3}$. This avoids treating a favorable individual disturbance realization as evidence that a finite-preview policy outperforms the optimum.

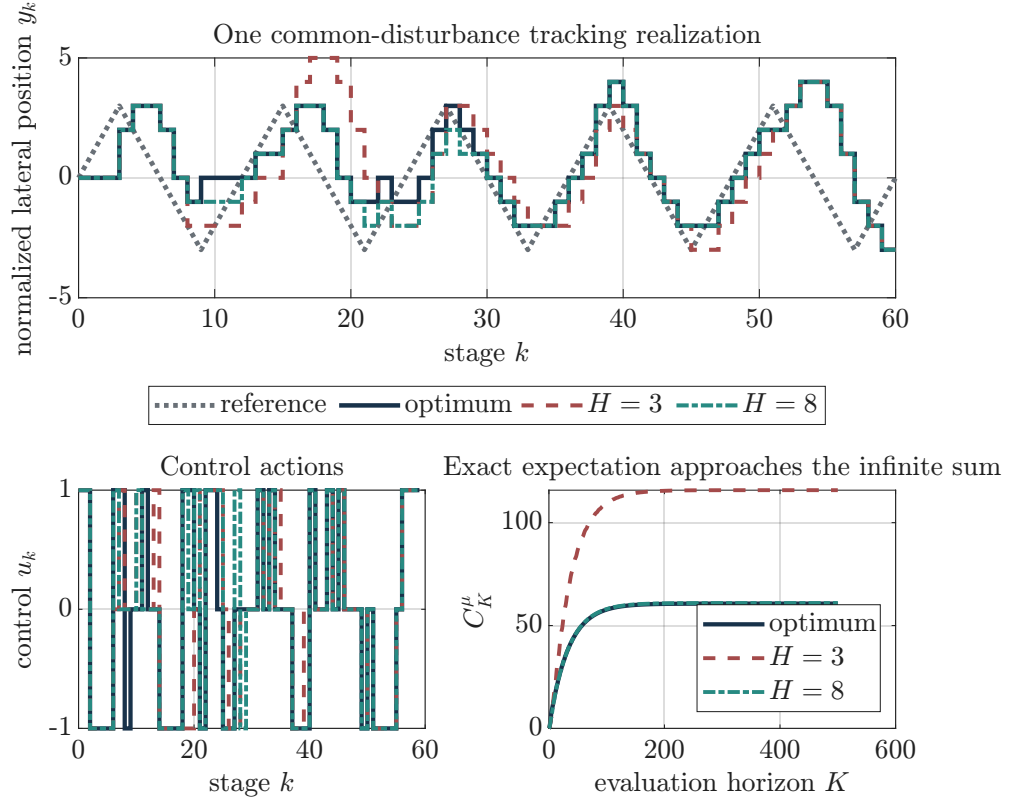


Figure 2.22: Behavior and performance of the exact periodic policy and finite-preview policies with $H = 3$ and $H = 8$. Tracking and controls use one common disturbance realization; the expected cumulative-cost panel is computed by exact probability propagation.

2.11 Summary

Chapter 2 established the exact reference for the book's learning-based methods: the Bellman recursion obtained when every required state, conditional expectation, and minimization is represented and evaluated exactly.

Table 2.6: Four DP reference structures and roles carried forward from Chapter 2.

Structure	Bellman object and computation	Main interpretation
Finite horizon	J_k^* and μ_k^* by backward DP	Stage-indexed value and policy with a terminal boundary; the model may also depend on stage
Stationary discounted problem	J^* and μ^* by VI or PI	One reusable Bellman fixed point; $0 < \alpha < 1$ gives contraction
Stage-augmented finite SSP	$\bar{J}^*$ and $\bar{\mu}^*$ by undiscounted VI or PI with $\bar{J}(\Delta) = 0$	Exact finite-horizon representation; convergence follows from acyclic finite termination
Approximate DP	Approximate problem or Bellman component using fitting, sampling, surrogate optimization, or rollout	Declare the approximation site and validate the resulting closed loop

Bellman's principle lets DP reuse cost-to-go subproblems instead of enumerating complete control sequences. Stochastic DP first requires declared information timing and a controlled Markov state, and exact DP then needs the relevant transition-and-cost model at every state and feasible control. Q-factors admit sample estimates but are not intrinsically model-free.

The HEV study uses physical structure to approximate a large energy-management OCP and a small grid to expose exact backward DP. The slalom study shows that stage augmentation preserves a finite problem exactly, whereas periodic continuation defines a new infinite-horizon objective. Its horizon sweep also distinguishes the finite-horizon planning value $J_{0|H}^*$ used to construct μ_H from the continuing policy value J_{μ_H} achieved by repeatedly applying that law.

What carries forward.

Chapter 3 asks how values, Q-factors, policies, and models can be represented without exact tables. Chapter 4 develops fitted and sample-based Bellman computations, approximate PI, rollout, and online lookahead. The audit questions remain: what is the objective, what makes the state Markov, what is approximated, what model information is used, and how is the closed loop evaluated?

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